

Design and Comparison of Buffers for Improving Load Transient Performance of Low-DropOut (LDO) Regulator

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I. INTRODUCTION

Background & Motivation

1. Gate-Dominant Pole LDO Regulator

- Conventional LDO regulators employ a large pass transistor to achieve low dropout voltage and support high load current.
- In on-chip LDOs with small load capacitors, the large parasitic capacitance at the pass transistor gate node tends to form the dominant pole.
- This gate-dominant pole limits loop bandwidth and degrades load transient response and PSR performance.

2. Necessity of Buffer Stage

- A buffer stage inserted between the Error Amplifier (EA) and pass transistor gate improves gate-driving capability.
- Enhanced driving strength increases loop bandwidth and improves transient response.

3. Buffer Requirements

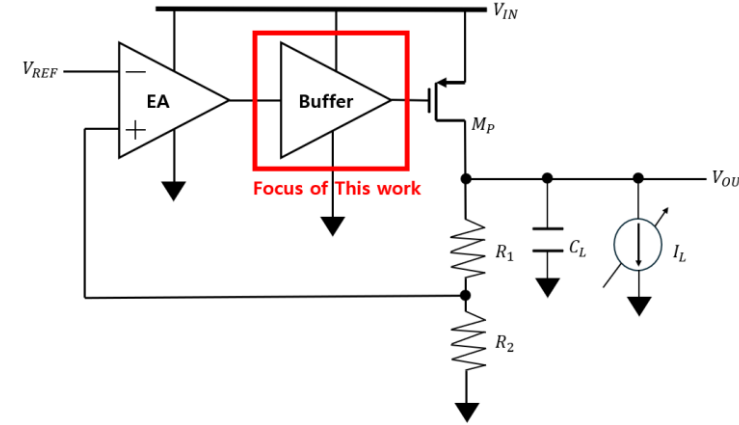
(Basic) Low Output Impedance

- Reduced gate pole effect, thereby loop bandwidth is extended

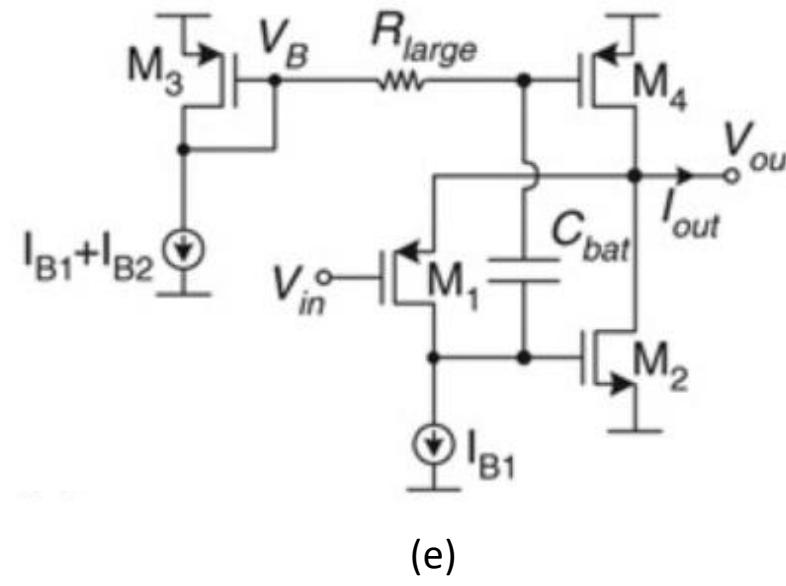
(Advanced) Push-Pull / Transient-Enhanced Operation

- Efficient current sourcing and sinking during rapid load transients.
- Fast charging and discharging of the pass transistor gate
- Improved transient response under abrupt load variations

To evaluate the impact of buffer topology on transient response, we designed and implemented three different buffer structures between the same Error Amplifier (EA) and pass transistor.



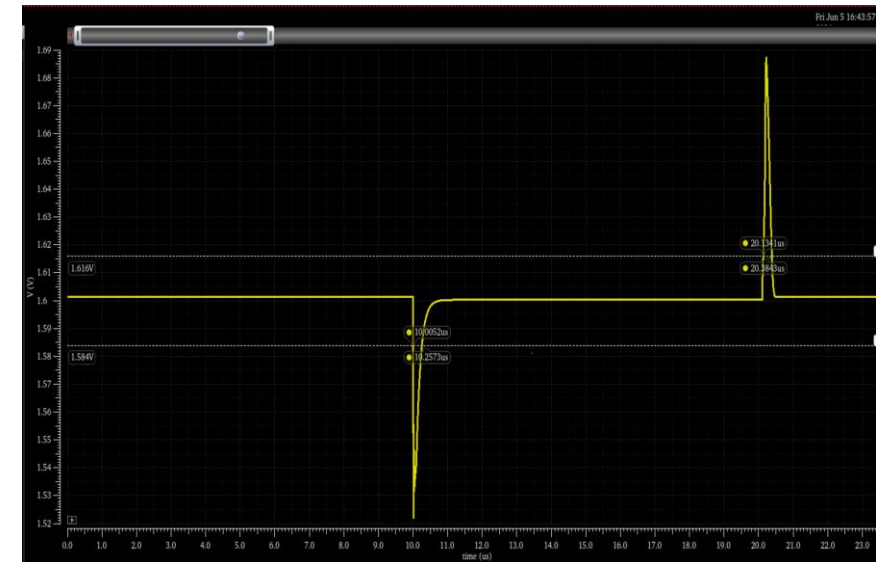
Class-AB Super Source Follower (SSF)



Main Characteristics of Class-AB SSF

- The Class-AB SSF implemented of combining the simple voltage buffering and gate driving capability of the Super Source Follower(SSF) with the low output impedance and fast local feedback characteristics of the Flipped-Voltage Follower(FVF).
- M1-M2 form a local feedback loop, which rapidly compensates for output-voltage variations and reduces the output impedance.
- R_{large} sets the DC bias of M4, while C_{bat} transfers transient voltage variations to M4, enabling Class-AB push-pull operation.
- The push-pull stage provides large charging and discharging currents during transients, achieving fast slew rate and settling time with low quiescent current.

III. SIMULATION RESULTS



Current 1	50u A
Current 2	10m A
Period	30u s
Delay time	
Rise time	100n s
Fall time	100n s
Pulse width	10u s

(d)

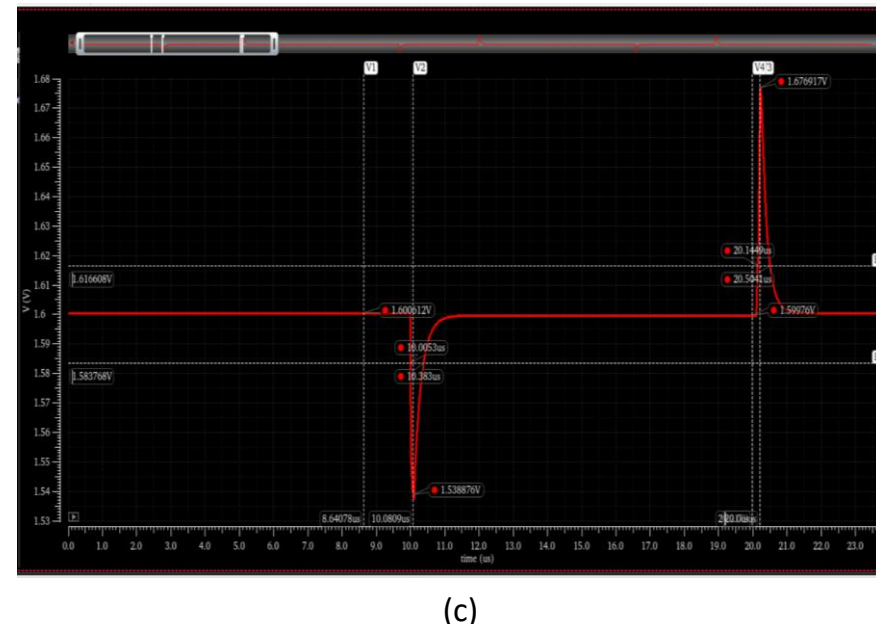
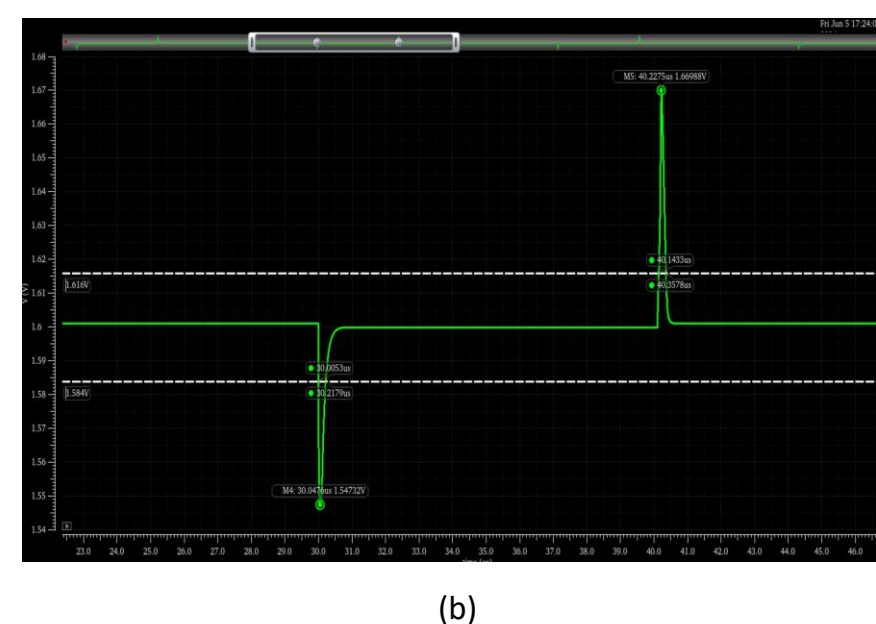


Fig 3. (a) Simulation results by using Source Follower (b) using Super Source Follower (c) using LTE buffer (d) Load-current pulse conditions for transient response simulation

Common Design Condition

V_{IN}	1.8V
Dropout Voltage	$\approx 200\text{mV}$
Load Capacitor C_L	80pF
Pass Transistor Size (W/L)	250um/65nm (Width = 1um X Finger 250)
Feedback Network (Resistive Divide)	$R_1 \approx 100\text{k}\Omega$ $R_2 \approx 100\text{k}\Omega$

Table I. Common Design Conditions

$$FoM = T_{settle} \times \Delta V_{out} \times I_Q$$

T_{settle} = undershoot/overshoot settling time
 ΔV_{out} = undershoot/overshoot voltage amplitude
 I_Q = Quiescent Current of the Buffer Stage

	Simple Source Follower	Class-AB SSF	LTE Buffer
I_Q of Buffer Stage(uA)	58.92	38	43.302
Phase Margin	@ Light load: 84.175° @ Heavy load: 54.1°	85.91° 72.928°	96° 59.519°
Overshoot amplitude(mV)	86	69.8	77.157
Undershoot amplitude(mV)	78	52.7	61.736
1% Settling time after overshoot (ns)	384.3	214.5	504.1
1% Settling time after undershoot (ns)	257.3	213.6	383
FoM (Figure of Merit) ($\times 10^{-12}$)	Overshoot: 1.947 Undershoot: 1.182	0.569 0.428	1.684 1.024
Unity Gain Frequency	@ Light load: 4.9503 MHz @ Heavy load: 63.712 MHz	11.4 MHz 380.5 MHz	6.1224 MHz 116.28 MHz

Table II. Simulation Measurement Results

IV. CONCLUSIONS

- Designed and simulated using TSMC 65-nm process under the same LDO operating conditions.
- Simple SF has a simple structure and stable operation, making it suitable as a baseline buffer structure. However, its limited gate-driving capability results in relatively limited transient improvement. It shows larger undershoot/overshoot and its bandwidth extension and FoM are relatively limited compared with the other two buffer structures.
- The LTE buffer employs an AC-coupling-based transient current-boosting mechanism to provide additional charging and discharging current to the pass transistor gate during rapid load transients. As a result, the gate voltage can be adjusted more effectively during the initial transient period, leading to smaller overshoot and undershoot voltages compared to the source follower (SF) buffer.
- However, the boosting paths are activated only during transient events and gradually lose their effect as the output approaches the steady-state condition. Consequently, the final settling behavior is primarily governed by the main feedback loop and the recovery of internal bias nodes.
- In addition, the AC-coupling network introduces large RC recovery time constants that may slow the recovery of internal nodes toward their nominal bias conditions, forming a low-frequency pole-zero doublet that results in a slow recovery tail. Therefore, although the LTE buffer effectively suppresses transient voltage deviations and achieves superior bandwidth enhancement and FoM performance, it may exhibit a longer settling time than the SF buffer.

II. THEORETICAL BACKGROUND

Buffer Output Impedance Reduction Techniques

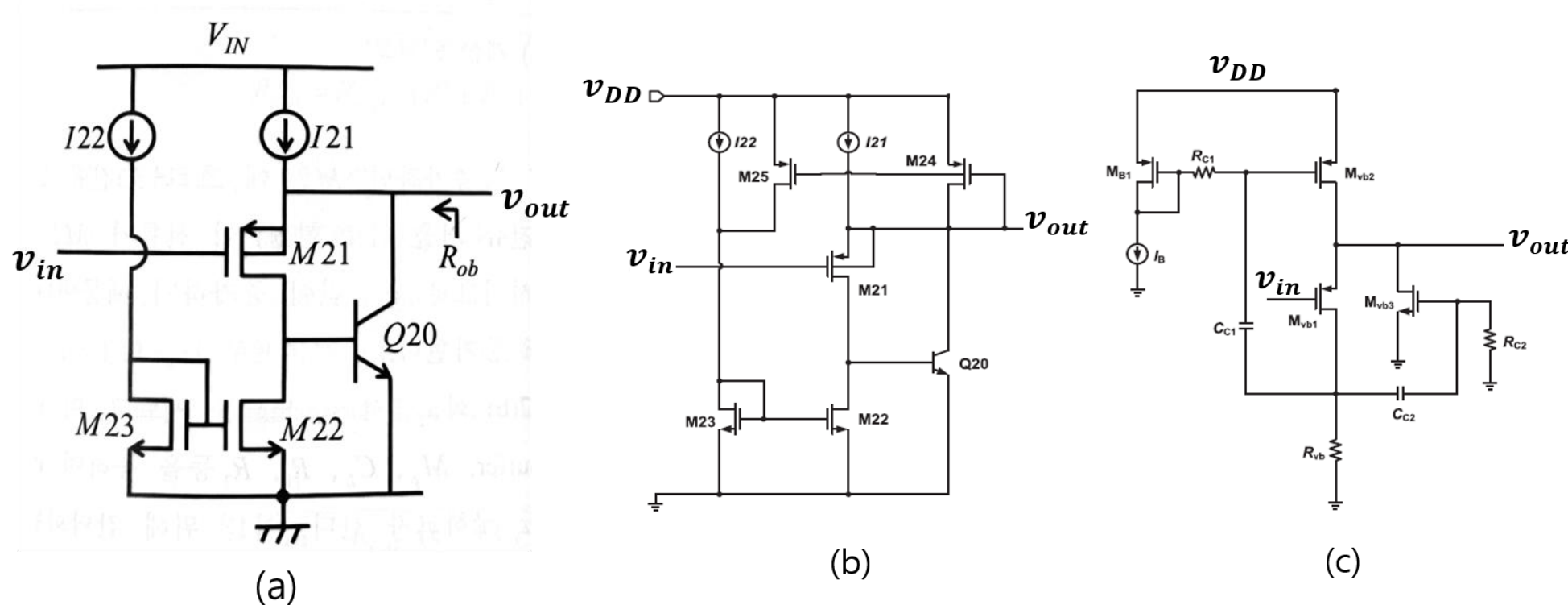
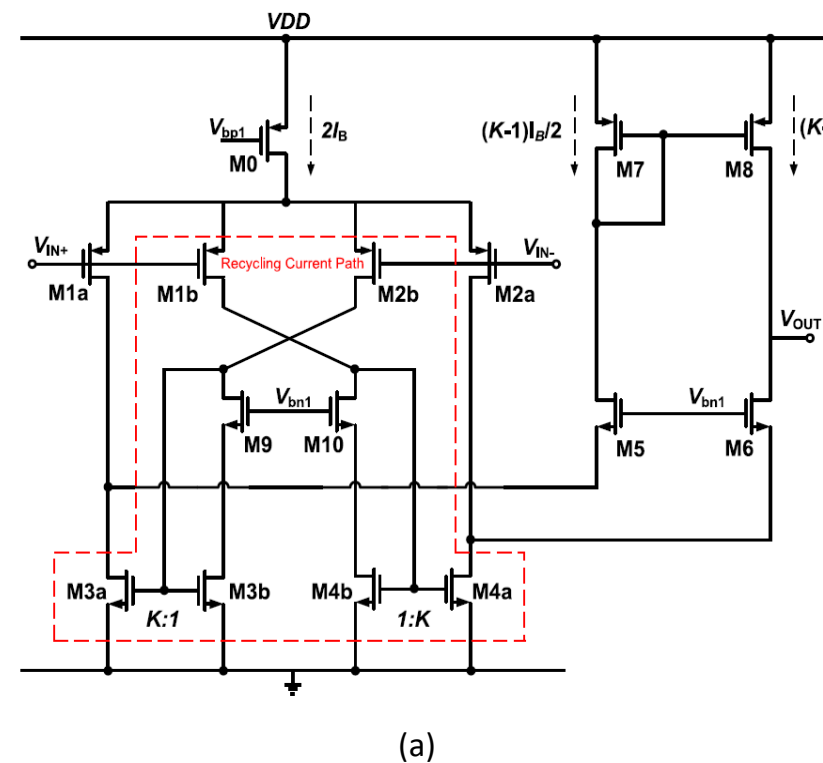


Fig 1. (a) Current Shunt feedback (b) Dynamic current-biasing shunt-feedback (c) with current-boosting circuit

Key Building Blocks

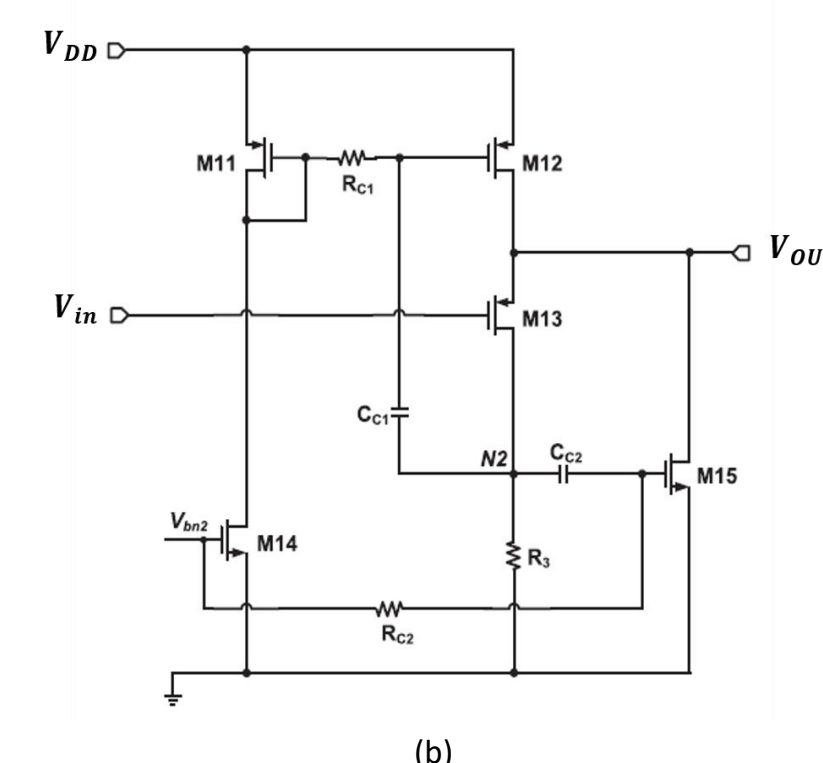
Recycling Folded Cascode (RFC) Amplifier



Main Characteristic of RFC Amplifier

- The input differential pair is implemented using split transistors.
- M3a, M3b, M4a, M4b : Serves as a biasing device and participates in small-signal amplification. (current recycling)
- Cross-coupling causes the small-signal currents from both branches to combine in-phase at the output node.
- Provides higher G_m , R_{out} , gain, UGF, and Slew Rate while maintaining the same power consumption as a conventional folded-cascode amplifier.

Low-Impedance Transient-current Enhanced (LTE) Buffer



Main Characteristic of LTE Buffer

- The LTE buffer is designed by combining current-shunt feedback and dynamic current boosting technique.
- Two AC-coupling paths are employed to provide additional charging and discharging current for the pass transistor gate only during rapid load transients, while maintaining low quiescent current and low output impedance.
- M11 and M12 form a PMOS current mirror for bias generation.
- M14-M15 form an NMOS current mirror for bias generation.
- The M12 and M15 are designed to be large.
- The R_3 resistor is designed to be small.
- The R_{C1} , R_{C2} resistor is designed to be large relatively.
- When the load current increases, M12 turns off while M15 turns on strongly, providing additional discharge current.
- When the load current decreases, M15 turns off while M12 turns on strongly, providing additional charge current.

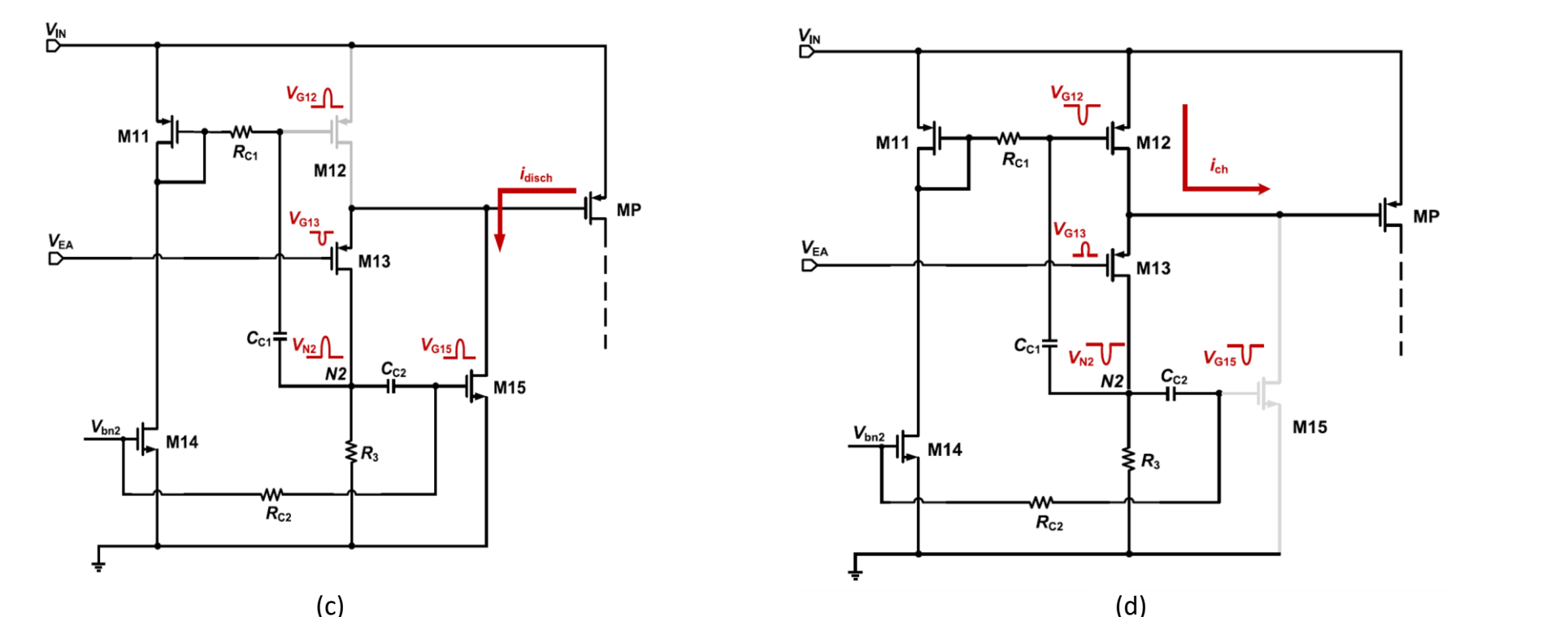


Fig 2. (a) Schematic of RFC Amplifier (b) Schematic of LTE Buffer (c) LTE Buffer Operation for a Load Current Increase (d) LTE Buffer Operation for a Load Current decrease (e) Schematic of Class-AB Super Source Follower

References

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